

Lab Experiment: 2

Title: Dataset Creation and Data Warehouse Star Model Design

Course title: **Data Mining Problem Description**

Course code: CSE-452

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Below is the **SQL script** to create the Star Schema:

- Dimension Tables
- Fact Table
- Primary Keys (PK)
- Foreign Keys (FK)
- Sample Data Inserts

This is written in standard **MySQL / PostgreSQL compatible SQL**.

◆ Create Database

```
CREATE DATABASE ThesisDW;  
  
USE ThesisDW;
```

◆ Create Dimension Tables

Student_Dimension

```
CREATE TABLE Student_Dimension (  
    Student_ID INT PRIMARY KEY,  
    CGPA DECIMAL(3,2),  
    Department VARCHAR(50),  
    Semester VARCHAR(20),  
    Attendance_Percentage DECIMAL(5,2)  
);
```

Teacher_Dimension

```
CREATE TABLE Teacher_Dimension (  
    Teacher_ID INT PRIMARY KEY,  
    Expertise_Area VARCHAR(100),
```

```
Evaluation_Rating DECIMAL(3,2),  
Supervised_Groups INT  
);
```

Thesis_Dimension

```
CREATE TABLE Thesis_Dimension (  
    Thesis_ID INT PRIMARY KEY,  
    Thesis_Topic VARCHAR(200),  
    Research_Domain VARCHAR(100),  
    Thesis_Type VARCHAR(50)  
);
```

Time_Dimension

```
CREATE TABLE Time_Dimension (  
    Time_ID INT PRIMARY KEY,  
    Month VARCHAR(20),  
    Semester VARCHAR(20),  
    Year INT  
);
```

Create Fact Table

Thesis_Performance_Fact

```
CREATE TABLE Thesis_Performance_Fact (  
    Fact_ID INT PRIMARY KEY,  
    Student_ID INT,  
    Teacher_ID INT,  
    Thesis_ID INT,  
    Time_ID INT,  
    Final_Marks DECIMAL(5,2),  
    Submission_Days INT,  
    Feedback_Score DECIMAL(3,2),  
    Task_Completion_Percentage DECIMAL(5,2),  
  
    FOREIGN KEY (Student_ID) REFERENCES Student_Dimension(Student_ID),  
    FOREIGN KEY (Teacher_ID) REFERENCES Teacher_Dimension(Teacher_ID),  
    FOREIGN KEY (Thesis_ID) REFERENCES Thesis_Dimension(Thesis_ID),  
    FOREIGN KEY (Time_ID) REFERENCES Time_Dimension(Time_ID)  
);
```

◆ Insert Sample Data

Student Data

```
INSERT INTO Student_Dimension VALUES  
(1, 3.75, 'CSE', 'Spring 2024', 92.50),  
(2, 3.60, 'EEE', 'Spring 2024', 88.00),  
(3, 3.90, 'CSE', 'Fall 2024', 95.00);
```

Teacher Data

```
INSERT INTO Teacher_Dimension VALUES
```

```
(101, 'Machine Learning', 4.50, 5),
```

```
(102, 'Power Systems', 4.20, 3),
```

```
(103, 'Data Science', 4.80, 6);
```

Thesis Data

```
INSERT INTO Thesis_Dimension VALUES
```

```
(201, 'AI Based Disease Prediction', 'Artificial Intelligence', 'Group'),
```

```
(202, 'Smart Grid Optimization', 'Electrical Engineering', 'Individual'),
```

```
(203, 'Big Data Analytics', 'Data Science', 'Group');
```

Time Data

```
INSERT INTO Time_Dimension VALUES
```

```
(301, 'June', 'Spring', 2024),
```

```
(302, 'July', 'Spring', 2024),
```

```
(303, 'December', 'Fall', 2024);
```

Fact Table Data

```
INSERT INTO Thesis_Performance_Fact VALUES
```

```
(1001, 1, 101, 201, 301, 85.50, 120, 4.60, 95.00),
```

```
(1002, 2, 102, 202, 302, 78.00, 130, 4.20, 88.50),
```

```
(1003, 3, 103, 203, 303, 91.75, 110, 4.80, 98.00);
```

Below is the sample output (SELECT results) for each table based on the sample data I previously provided.

SELECT * FROM table_name;

◆ **1** Student_Dimension

SELECT * FROM Student_Dimension;

 Output:

Student_ID	CGPA	Department	Semester	Attendance_Percentage
1	3.75	CSE	Spring 2024	92.50
2	3.60	EEE	Spring 2024	88.00
3	3.90	CSE	Fall 2024	95.00

◆ **2** Teacher_Dimension

SELECT * FROM Teacher_Dimension;

 Output:

Teacher_ID	Expertise_Area	Evaluation_Rating	Supervised_Groups
101	Machine Learning	4.50	5
102	Power Systems	4.20	3
103	Data Science	4.80	6

◆ **3** Thesis_Dimension

SELECT * FROM Thesis_Dimension;

 Output:

Thesis_ID	Thesis_Topic	Research_Domain	Thesis_Type
201	AI Based Disease Prediction	Artificial Intelligence	Group
202	Smart Grid Optimization	Electrical Engineering	Individual
203	Big Data Analytics	Data Science	Group

◆ Time_Dimension

SELECT * FROM Time_Dimension;

 Output:

Time_ID	Month	Semester	Year
301	June	Spring	2024
302	July	Spring	2024
303	December	Fall	2024

◆ Thesis_Performance_Fact

SELECT * FROM Thesis_Performance_Fact;

 Output:

Fact_ID	Student_ID	Teacher_ID	Thesis_ID	Time_ID	Final_Marks	Submission_Days	Feedback_Score	Task_Completion_Percentage
1001	1	101	201	301	85.50	120	4.60	95.00
1002	2	102	202	302	78.00	130	4.20	88.50
1003	3	103	203	303	91.75	110	4.80	98.00

◆ Joined Star Schema Output (Full Report View)

If you run this:

SELECT

s.Student_ID,

s.Department,

t.Expertise_Area,

th.Thesis_Topic,

tm.Month,

tm.Year,
f.Final_Marks,
f.Feedback_Score

FROM Thesis_Performance_Fact f
JOIN Student_Dimension s ON f.Student_ID = s.Student_ID
JOIN Teacher_Dimension t ON f.Teacher_ID = t.Teacher_ID
JOIN Thesis_Dimension th ON f.Thesis_ID = th.Thesis_ID
JOIN Time_Dimension tm ON f.Time_ID = tm.Time_ID;

 Output:

Student_ID	Department	Expertise_Area	Thesis_Topic	Month	Year	Final_Marks	Feedback_Score
1	CSE	Machine Learning	AI Based Disease Prediction	June	2024	85.50	4.60
2	EEE	Power Systems	Smart Grid Optimization	July	2024	78.00	4.20
3	CSE	Data Science	Big Data Analytics	December	2024	91.75	4.80